

Diff EQ

Section 6.2

12/1/25

A) Differential Equation

Ex: $\frac{dy}{dx} = \frac{2x}{y}$ $f(1) = -3$

$$y dy = 2x dx$$

$$\int y dy = \int 2x dx$$

step 1

$$\frac{1}{2} y^2 = x^2 + C$$

$$\frac{1}{2} (-3)^2 = 1 + C$$

$$C = \frac{7}{2}$$

$$\frac{1}{2} y^2 = x^2 + \frac{7}{2}$$

$$y^2 = 2x^2 + 7$$

step 2

$$y = \pm \sqrt{2x^2 + 7}$$

$$y = -\sqrt{2x^2 + 7}$$

Ex: $y' = x^2 \sqrt{y}$ $y(2) = 4$

$$\int y^{\frac{1}{2}} dy = \int x^2 dx$$

$$2y^{\frac{1}{2}} = \frac{1}{3} x^3 + C$$

step 1

$$2(4) = \frac{1}{3} (8) + C$$

$$-5 = C$$

$$2y^{\frac{1}{2}} = \frac{1}{3} x^3 - 5$$

$$y^{\frac{1}{2}} = \frac{1}{6} x^3 - \frac{5}{2}$$

$$y = \left(\frac{1}{6} x^3 - \frac{5}{2} \right)^2$$

Ex: $y' = x^2 \sqrt{y}$ $y(2) = 4$

B) Do it with

Ex: $\frac{dy}{dx} = 9x^2 y$ and $y(0) = -2$

$$\int \frac{1}{y} dy = \int 9x^2 dx$$

$$\ln|y| = 3x^3 + C$$

$$e^{\ln|y|} = e^{3x^3 + C}$$

$$e^{\ln|y|} = e^{3x^3} \cdot e^C$$

$$e^{\ln|y|} = C e^{3x^3}$$

$$-y = C e^{3x^3}$$

$$y = C e^{3x^3}$$

$$-2 = C e^0$$

$$-2 = C$$

$$y = -2 e^{3x^3}$$

Ex: $y' = \frac{y+3}{x^2}$, $x \neq 0$, and $y(2) = 1$

$$\frac{1}{y+3} dy = \frac{1}{x^2} dx$$

$$\ln|y+3| = -\frac{1}{x} + C$$

$$e^{\ln|y+3|} = e^{-\frac{1}{x} + C}$$

$$y+3 = C e^{-\frac{1}{x}}$$

$$y+3 = C e^{-\frac{1}{x}}$$

$$y = C e^{-\frac{1}{x}} - 3$$

$$u = C e^{-\frac{1}{x}}$$

$$\frac{4}{e^{-\frac{1}{2}}} = C e^{\frac{1}{2}} = C$$

$$y = 4 e^{\frac{1}{2} - \frac{1}{x}} - 3$$

$\frac{dy}{dx} = 4x^3 y + 20x^3$ and $y(1) = -3$

$$\frac{dy}{dx} = 4x^3 (y+5)$$

$$\int \frac{1}{y+5} dy = \int 4x^3 dx$$

$$\ln|y+5| = x^4 + C$$

$$e^{\ln|y+5|} = e^{x^4 + C}$$

$$y+5 = C e^{x^4}$$

$$2 = C e^1$$

$$2 = C e^1$$

$$C = \frac{2}{e}$$

$$y+5 = \frac{2}{e} e^{x^4}$$

$$y = 2 e^{x^4-1} - 5$$